

Purity of the completed root state

`MathlibAnnex.CStarAlgebra.CAR.isPureState_rootState`

theorem

Passes extremality of the root vector states from every finite matrix stage to the completed CAR algebra.

Statement

Let A be the completed CAR algebra with stage embeddings $\iota_n : M_{2^n}(\mathbb{C}) \rightarrow A$. Let $\omega : A \rightarrow \mathbb{C}$ be the continuous linear functional characterized by $\omega(\iota_n(a)) = a_{00}$ for every stage matrix a . Then ω is a pure state: it is an extreme point of the convex set of continuous positive unital complex-linear functionals on A .

Assumptions

The embeddings come from $a \mapsto a \otimes I_2$. The index 0 is the distinguished first coordinate in each stage. The coordinate functionals are compatible and bounded by the matrix norm, so they extend to the specified functional ω on the completion. Positivity and $\omega(1) = 1$ are established before the purity argument. Convex combinations use real coefficients.

Conclusion

If φ and ψ are states on A , $0 < t < 1$, and $\omega = t\varphi + (1-t)\psi$, then $\varphi = \psi = \omega$. Thus completion has not introduced a nontrivial convex decomposition of this state.

The purity asserted here is extremality in the state space. The proof uses the pure finite-stage restrictions and density; it makes no assertion that all states on A are pure.

Proof route

Restrict a proposed convex decomposition of ω to each finite matrix stage. The restriction of a state is again a state, because ι_n is unital and preserves positivity. The root coordinate state $a \mapsto a_{00}$ is pure on each matrix algebra, so both restricted states equal it. The two continuous functionals therefore agree with ω on the dense union of stages, and hence on all of A .

▼ Proof steps

1. Write $\omega = t\varphi + (1-t)\psi$ with states φ, ψ and $0 < t < 1$.
2. For each n , compose with ι_n to obtain a convex decomposition of $a \mapsto a_{00}$ into states on $M_{2^n}(\mathbb{C})$.
3. Finite-stage purity forces $\varphi \circ \iota_n = \psi \circ \iota_n = (a \mapsto a_{00})$.
4. Continuity and density imply $\varphi = \psi = \omega$. The source uses the equivalent one-sided extreme-point criterion, establishing the first equality, which suffices.

Main citations

- Exact declaration and proof
- MathlibAnnex.CStarAlgebra.CAR.rootState
- MathlibAnnex.CStarAlgebra.CAR.rootState_stage
- MathlibAnnex.CStarAlgebra.CAR.rootState_mem_stateSpace
- MathlibAnnex.CStarAlgebra.CAR.restrictState_mem_stateSpace
- MathlibAnnex.CStarAlgebra.CAR.isPureState_rootFunctional
- MathlibAnnex.CStarAlgebra.CAR.eq_rootState_of_restrict
- MathlibAnnex.CStarAlgebra.CAR.dense_stageRange

Lean source signature (exact)

theorem isPureState_rootState : MathlibAnnex.CStarAlgebra.IsPureState Limit rootState

In the source	Mathematical meaning
Limit; rootState	$\mathcal{A}$, the completed CAR algebra, and ω , the state whose value on $\iota_n(a)$ is a_{00} .
MathlibAnnex.CStarAlgebra.IsPureState Limit rootState	The conclusion that ω belongs to the extreme points of the state space of $\mathcal{A}$, viewed as a real convex set.
IsPureState	Here the state space consists of continuous complex-linear functionals φ positive on positive elements with $\varphi(1) = 1$. Extremality says that $\omega = t\varphi + (1 - t)\psi$, with states φ, ψ and $0 < t < 1$, forces $\varphi = \psi = \omega$. There is no state or representation input to this fixed-algebra theorem.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.CStarAlgebra.CAR.isPureState_rootState
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Accepted source guide SHA-256: c75ad15fb26a412c334d8294d39b6a38e0ee67de5e8db8984085c723d94ff74f
Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73